

# LLM-Mediated Human-AI Interaction in Search and Rescue: Impact of Expertise on Attentional Allocation

Human performance, visual attention, and expertise under LLM guidance

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# Human-AI Setting in **Critical** Situation

Operators must locate and recover people while conditions change quickly.

- Missions occur in **challenging environments**
- Decisions must be made under **time pressure**
- Need **reliable partner** to help humans <sup>1</sup>



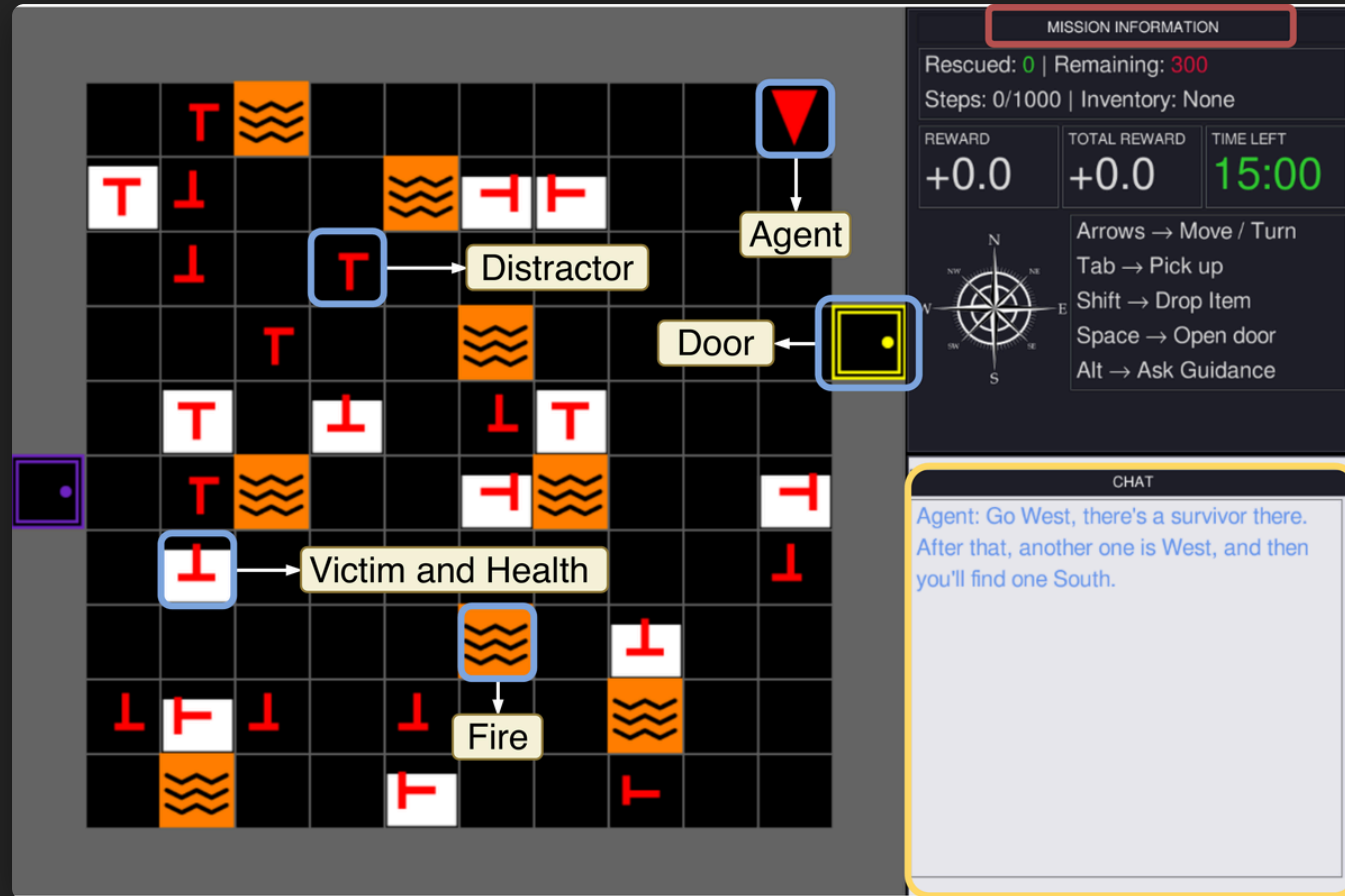
# Research Questions and **Contribution**

- **Performance:** Does LLM guidance improve task outcomes?
- **Attention:** How does guidance change visual attention and eye-tracking metrics?
- **Expertise:** Does expertise moderate these effects?
- **Testbed:** A controlled simulation supports repeatable human-AI teaming research.



We study how humans and AI **process information together**, using **eye tracking** to inform **adaptive AI** for the future.

# Study Design In Brief



**13**

PARTICIPANTS

**3**

TRIALS EACH (RANDOMIZED)

**2 + 1**

LLM + BASELINE

**15 min**

OR 1,000 MOVES

**Task:** rescue victims efficiently in a partially observable grid while victim health deteriorates.

**ALT** requests advice and reveals victim health for **5 seconds**.



# Expertise Level



- **Expertise labeling:** K-means clustering ( $K = 2$ ) grouped participants into expert vs. novice<sup>2</sup>

# How The LLM Was Prompted

## **GUIDANCE STYLE**

- You are a search-and-rescue coordinator giving calm, human-like radio guidance to the operator.
- Instructions must be directional and action-focused, always using North, South, East, West, with no coordinates, step counts, or visible reasoning.
- Output either a sparse one-sentence instruction or a detailed 3–4 sentence radio briefing, always concise, natural, and wrapped strictly as <START> ... <END>.

## **PRIORITY**

- Prioritize visible reachable survivors who are still saveable, nearest first.
- From the current state {obs}, choose one immediate objective by prioritizing the nearest reachable visible saveable survivor, then the next saveable survivor if relevant.
- Otherwise choose the best remaining alive victim, then the nearest reachable survivor or key, or finally explore visible areas that may open a new path.

## **MISSION RULES**

- Follow mission and environment rules: rescue all survivors.
- Use matching keys for locked doors, unlocked doors can be opened, carry only one key, avoid lava.
- Never mention unreachable objects. Final instruction must be rule-compliant.

- **Grid-based world:** each cell represents an object such as empty space, wall, victim, lava, door, or key. <sup>3</sup>
- **Reachability check:** BFS determines which targets are reachable and their distance from the player. <sup>4</sup>
- Models: **OpenAI** and **Gemini**.



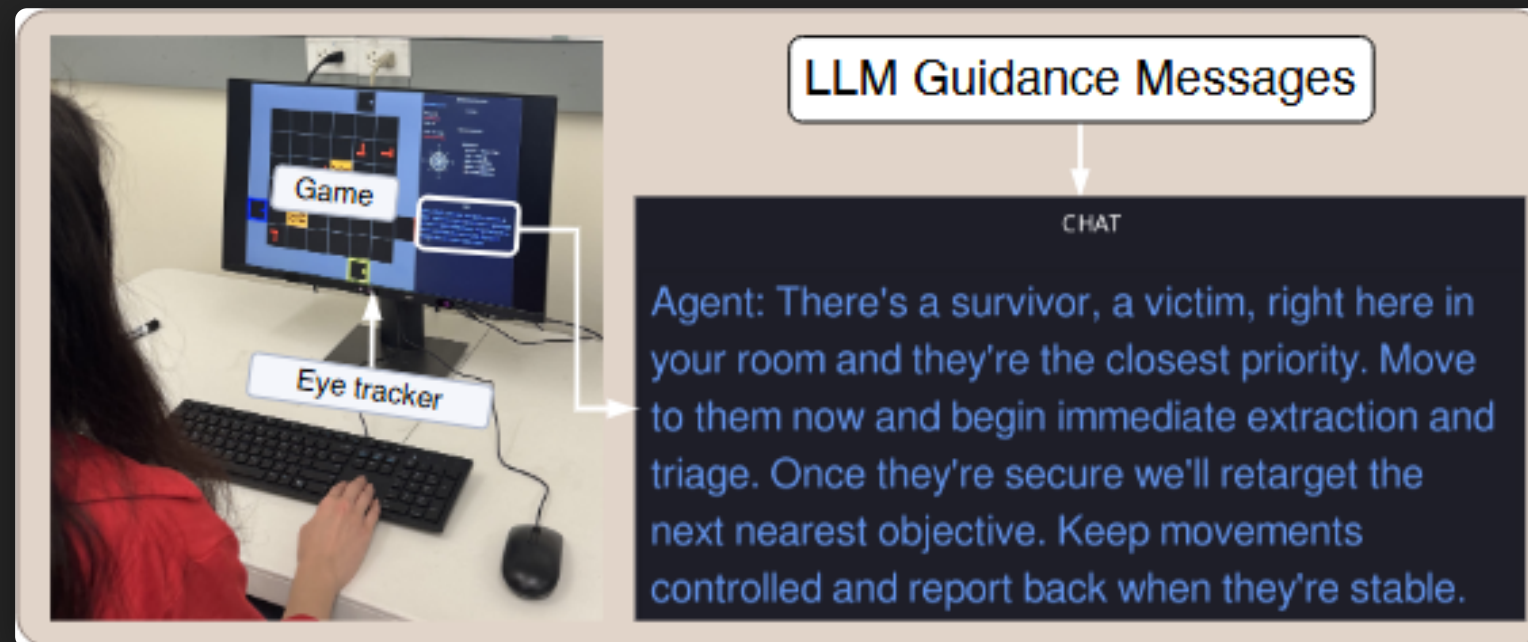
# Task and Eye-Tracking **Features**

| FEATURE                  | WHAT IT SHOWS                                |
|--------------------------|----------------------------------------------|
| <b>Victims per step</b>  | Rescue efficiency over time                  |
| <b>Total reward</b>      | Overall task performance                     |
| <b>Saved victims</b>     | Mission success                              |
| Fixation count           | How often the participant focused on an area |
| Fixation duration        | How long the participant looked at one area  |
| Saccade duration         | Duration of fast eye movements               |
| Pupil diameter SD        | Variation in pupil size, related to workload |
| Game-area fixation time  | Attention to the SAR environment             |
| Chat-panel fixation time | Attention to the LLM guidance                |

# Data Collection and Analysis

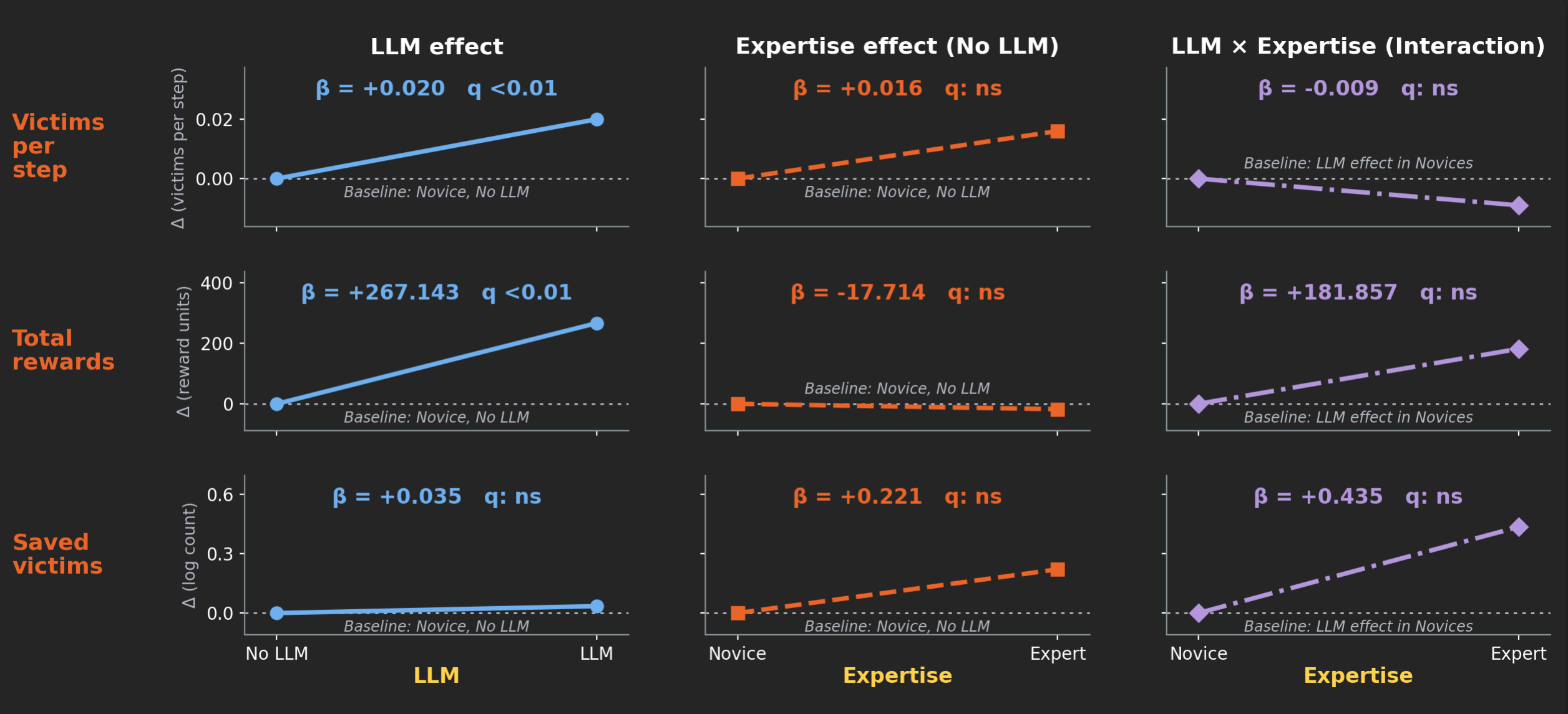
- Synchronized data streams via **LSL (Lab Streaming Layer)**
- Gaze and pupil data recorded with a **Tobii eye tracker**

**Mixed-effects model:** outcome = LLM condition + LLM × expertise + expertise + participant random intercept.

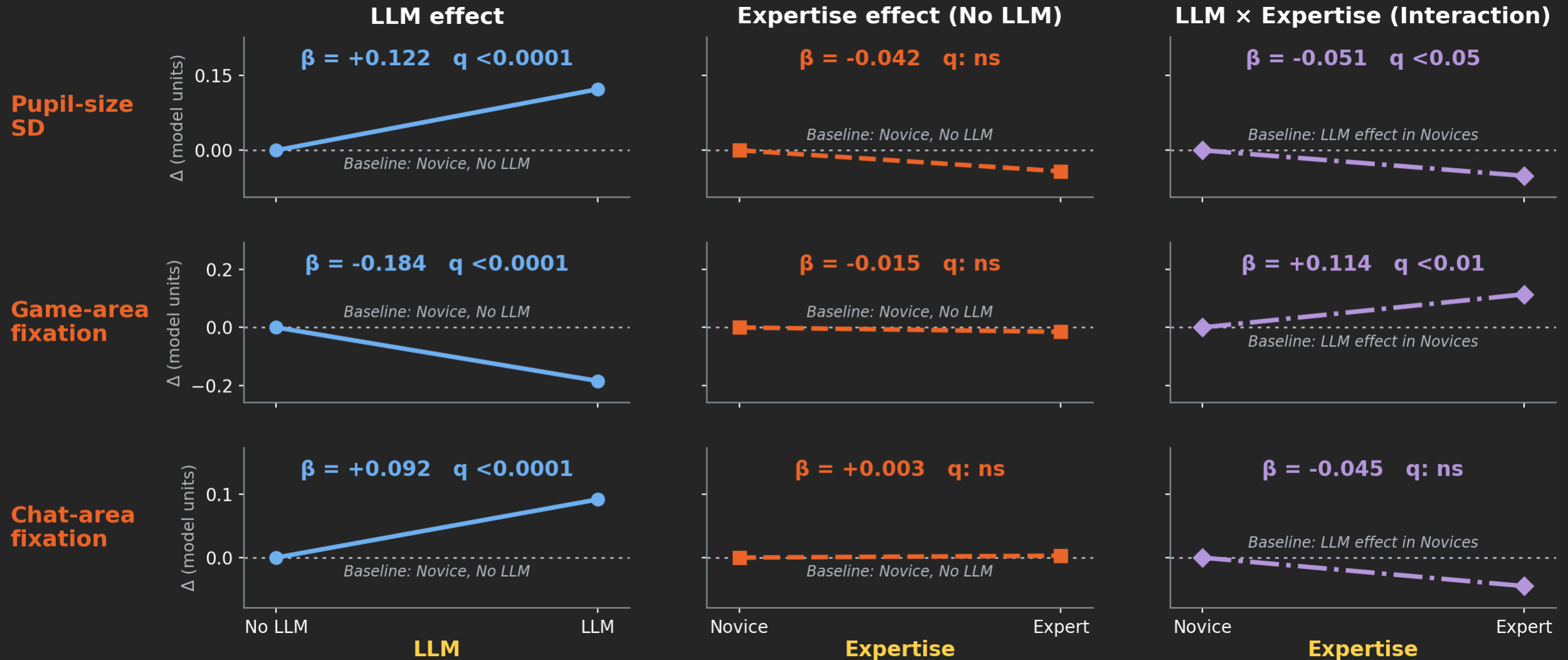




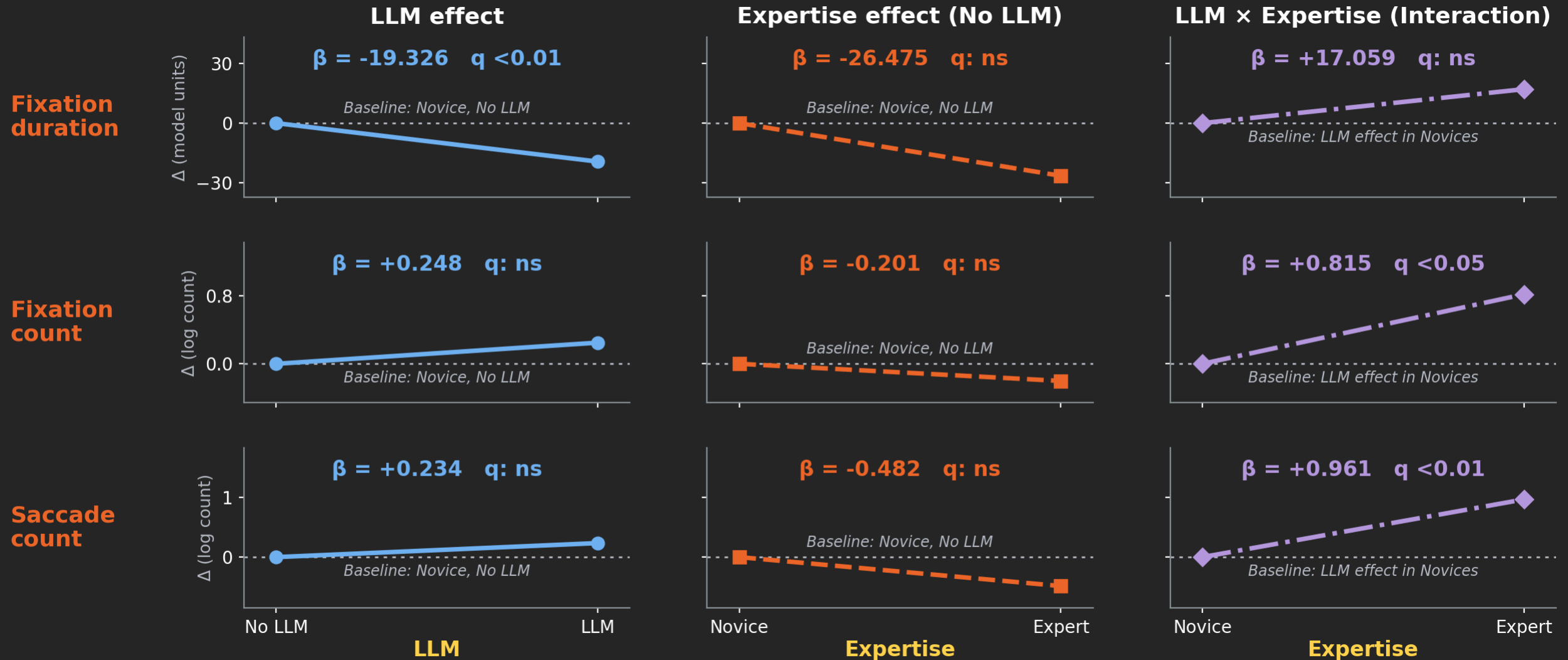
# Results: Performance



# Results: Pupil and Area Fixation



# Results: Fixation and Saccade





# Expertise Changes **How Guidance Is Used**

## **EXPERTS**

Maintained more engagement with the game under LLM support and showed stronger changes in visual scanning.

## **NOVICES**

Shifted more attention away from the game, consistent with greater reliance on the guidance panel.

**LLM guidance received**



### **Expert**

Cross-checks  
guidance against the  
environment (scans,  
verifies)

### **Novice**

Follows guidance  
directly

# Limitations and Future Work

## Limitations

- Sample size
- Simulated grid-world task
- Incorrect/conflicting LLM guidance was not tested
- Additional physiological measures were not included

## Future work

### TRUST & RELIABILITY

Test how LLM errors affect operator responses and trust.

### ADAPTIVE GUIDANCE

Tailor help using performance and physiological signals to reduce mental workload.

### SITUATION AWARENESS

Use this data to infer workload, situational awareness, and human-AI interaction patterns.

### OTHER DOMAINS

Apply this approach to other human-AI teaming domains beyond search and rescue.

The next step is moving from **static guidance** toward **reliable, adaptive human-AI teaming**.

# References

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- [2] J. P. Distefano, H. Manjunatha, S. Chowdhury, K. Dantu, D. Doermann and E. T. Esfahani, "Using Physiological Information to Classify Task Difficulty in Human-Swarm Interaction," *2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Melbourne, Australia, 2021, pp. 1198-1203, doi: [10.1109/SMC52423.2021.9658653](https://doi.org/10.1109/SMC52423.2021.9658653).
- [3] Chevalier-Boisvert, M., Dai, B., Towers, M., Perez-Vicente, R., Willems, L., Lahlou, S., Pal, S., Castro, P. S., & Terry, J. (2023). Minigrid & Miniworld: Modular & Customizable Reinforcement Learning Environments for Goal-Oriented Tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 36.
- [4] Siklóssy, Laurent, A. Rich, and Vesko Marinov. "Breadth-first search: some surprising results." *Artificial Intelligence* 4.1 (1973): 1-27.





Thank you

Questions?